# Seminar 4 Risk-based approach to regulating artificial intelligence technology: the case of the EU

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# Course Structure

1. Recap & Overview: theories of governing AI
2. AI definition in General
3. Overview of Risk-based Approach in EU AI Act
    1. Hierarchical Framework
    2. Different types and functions of AI
4. Type of AI in AI Act
    - Unacceptable AI
    - High-risk AI
    - Limited-risk AI
    - General-purpose AI
5. Reflections
6. Group Activity
7. Conclusion

# Part I – Recap & Overview of the course

# Recap of Last Seminar

**Theoretical Foundations of AI Regulation:**

1. Public Interest Theories
2. Private Interest Theories
3. Institutionalist Theories

**Challenges in AI Regulation:**

1. Polycentric Regulation
2. Hard Law & Soft Law

**New Global Trends:**

1. Pro-innovation Approach;
2. Risk-based approach;
3. Deregulation

## Group 2. Generative AI and Large Language Models

Generative AI can refer to systems that create new content such as text images and pictures ,it learns from data.Large Language AI Models are type of generative AI trained on massive text datasets.

# Part II – AI definition in General in AI Act

Useful Resource: https://artificialintelligenceact.eu/ai-act-explorer/

# AI definition in AI Act

## The scope of the EU AI Act

### Article 2: Scope

Date of entry into force: **2 February 2025**

According to: **Article 113(a)**

Inherited from: **Chapter I**

See here for a full implementation timeline.

SUMMARY +

1. This Regulation applies to:

(a) providers placing on the market or putting into service AI systems or placing on the market general-purpose AI models in the Union, irrespective of whether those providers are established or located within the Union or in a third country;

(b) deployers of AI systems that have their place of establishment or are located within the Union;

(c) providers and deployers of AI systems that have their place of establishment or are located in a third country, where the output produced by the AI system is used in the Union;

(d) importers and distributors of AI systems;

(e) product manufacturers placing on the market or putting into service an AI system together with their product and under their own name or trademark;

(f) authorised representatives of providers, which are not established in the Union;

(g) affected persons that are located in the Union.

# AI definition in AI Act

## The scope of the EU AI Act

- Article 2(1)(a) AI Act: providers placing **on the market** or putting into service **AI systems** or placing on the market **general-purpose AI models** in the Union,
- Irrespective of whether those providers are established or located within the Union or in a third country;
- **Providers**: develops an AI system or a general-purpose AI model; or places it on the market or puts the AI system into service (whether for free or not)
- **Deployers**: deploying or using AI systems, except for personal purpose;
- **Importers**: Located in EU that places AI system in the EU on behalf of other entity in third country
- **Distributors**: in the AI supply chain to make AI system available on EU market, other than provider or importer.
- **Placing on the market**: first making available of an AI system or a general-purpose AI model on the Union market;
- Affected persons??

# AI definition in AI Act

### When does the EU AI Act not apply?

- It does not apply to AI systems used for *military, defense, or national security purposes*;
- Used by *foreign public authorities* or *international organizations* for *law enforcement* and judicial cooperation, <u>*provided they protect individuals' rights*</u>;
- Does not apply to AI systems used for <u>*scientific research and development*</u>, or to AI systems not yet on the market;
- It does not affect existing EU laws on data protection, privacy, and confidentiality;
- It also does not apply to <u>*individuals using AI systems for personal, non-professional activities*</u>, or to AI systems released under free and open-source licenses, unless they are high-risk or fall under certain articles.

# AI definition in AI Act

## What is AI system in the EU AI Act?

- The AI Act does not apply to all systems, but only to those systems that fulfil the definition of an 'AI system' within the meaning of Article 3(1) AI Act. The definition of an AI system is therefore key to understanding the scope of application of the AI Act.
- Art 3(1), AI Act: 'AI system' means a **machine-based system** that is designed to operate with **varying levels of autonomy** and that may **exhibit adaptiveness** after deployment, and that, for **explicit or implicit objectives**, **<u>infers</u>**, **<u>from the input it receives</u>**, how to **generate outputs** such as **predictions, content, recommendations, or decisions** that can **<u>influence</u> physical or virtual environments**;
- The definition of an AI system adopts a lifecycle-based perspective encompassing two main phases: (1) the pre-deployment or 'building' phase of the system and (2) the post-deployment or 'use' phase of the system.

# AI definition in AI Act

**Seven elements of AI definition in AI Act:**

(1) a machine-based system; (2) that is designed to operate with varying levels of autonomy; (3) that may exhibit adaptiveness after deployment; (4) and that, for explicit or implicit objectives; (5) infers, from the input it receives, how to generate outputs (6) such as predictions, content, recommendations, or decisions (7) that can influence physical or virtual environments.

**1. Machine-based system:**

- AI systems are developed with and run on machines. The term 'machine' can be understood to include both the hardware and software components that enable the AI system to function.
- Hardware: physical elements of the machine, infrastructural level (processing units, memory, storage devices, networking units, and input/output interfaces) e.g., CPU, GPU, Memory (RAM)
- Software: computer code, instructions, programs, operating systems, and applications that handle how the hardware processes data and performs tasks. E.g., OS system, PPT, Database

# AI definition in AI Act

- All AI systems needs to be machine-based.
  - require machines to enable their functioning, such as model training, data processing, predictive modelling and largescale automated decision making.
- The entire lifecycle of AI systems relies on machines that can include many hardware or software components.
- ‘machine-based’, very broad coverage:
  - quantum computing systems, which significant departure from traditional computing systems;
  - also biological or organic systems, e.g., BCI

# AI definition in AI Act

## 2. Autonomy:

- ‘designed to operate with varying levels of autonomy’. Recital 12 of the AI Act: AI systems are designed to operate with ‘some **degree of independence of actions from human involvement** and of **capabilities** to operate **without human intervention**’.
- Autonomy & inference: the **inference capacity** of an AI system (i.e., predictions, content, recommendations, or decisions that can influence physical or virtual environments) is key;
- Central to the concept of autonomy is ‘human involvement’ and ‘human intervention’;
- ‘some degree of independence of action’ in recital 12 AI Act excludes systems:
  - Operate solely with full manual human involvement and intervention.
  - Human involvement and human intervention can be either direct, e.g. through manual controls,
  - Or indirect, e.g. though automated systems-based controls which allow humans to delegate or supervise system operations.

# AI definition in AI Act

## 3. Adaptiveness:

- ***Autonomy*** and ***Adaptiveness*** are distinct but closely related.
- Recital 12 AI Act clarifies that '*adaptiveness*' refers to self-learning capabilities, allowing the behaviour of the system to change while in use. The new behaviour of the adapted system may produce different results from the previous system for the same inputs.
- Ability to automatically learn, discover new patterns, or identify relationships in the data beyond what it was initially trained on.
- The use of the term 'may' in relation to this element of the definition indicates that a system may, but does not necessarily have to, possess adaptiveness or self-learning capabilities after deployment to constitute an AI system. It is not a decisive condition for determining whether the system qualifies as an AI system.

# AI definition in AI Act

## A Comprehensive Survey of Self-Evolving AI Agents

### A New Paradigm Bridging Foundation Models and Lifelong Agentic Systems

**Jinyuan Fang**[*1], **Yanwen Peng**[*2], **Xi Zhang**[*1], **Yingxu Wang**[3], **Xinhao Yi**[1], **Guibin Zhang**[4], **Yi Xu**[5], **Bin Wu**[6], **Siwei Liu**[7], **Zihao Li**[1], **Zhaochun Ren**[8], **Nikos Aletras**[2], **Xi Wang**[2], **Han Zhou**[5], **Zaiqiao Meng**[1✉]

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## A Survey of Self-Evolving Agents

### What, When, How, and Where to Evolve on the Path to Artificial Super Intelligence

**Huan-ang Gao**[γ†], **Jiayi Geng**[α†], **Wenyue Hua**[ϵ†], **Mengkang Hu**[ω†], **Xinzhe Juan**[σμ†], **Hongzhang Liu**[ξ†], **Shilong Liu**[α†], **Jiahao Qiu**[αδ†], **Xuan Qi**[γ†], **Qihan Ren**[σ†], **Yiran Wu**[ρ†], **Hongru Wang**[k†✉], **Han Xiao**[τ†], **Yuhang Zhou**[λ†], **Shaokun Zhang**[ρ†], **Jiayi Zhang**[π], **Jinyu Xiang**, **Yixiong Fang**[θ], **Qiwen Zhao**[ζ], **Dongrui Liu**[σ], **Cheng Qian**[β], **Zhenhailong Wang**[β], **Minda Hu**[τ], **Huazheng Wang**[η], **Qingyun Wu**[ρ], **Heng Ji**[β], **Mengdi Wang**[αδ✉]

# AI definition in AI Act

### 4. AI system objectives:

- **Explicit objectives:** clearly stated goals that are directly encoded by the developer into the system. For example, they may be specified as the optimisation of some cost function, a probability, or a cumulative reward.
- **Implicit objectives:** goals that are not explicitly stated but may be deduced from the behaviour or underlying assumptions of the system. These objectives may arise from the training data or from the interaction of the AI system with its environment.
- The objectives of the AI system may be different from the intended purpose of the AI system in a specific context (Recital 12 AI Act).
- **Intended purpose** is externally oriented and includes the context in which the system is designed to be deployed and how it must be operated. Indeed, according to Article 3(12) AI Act, the intended purpose of an AI system refers to the 'use for which an AI system is intended by the provider'.

# AI definition in AI Act

**5. Inferencing how to generate outputs using AI techniques:**

- Recital 12 AI Act: "[a] key characteristic of AI systems is their capability to infer." AI systems should be distinguished from "simpler traditional software systems or programming approaches and should not cover systems that are based on the rules defined solely by natural persons to automatically execute operations."
- ISO/IEC 22989 standard: defines inference 'as reasoning by which conclusions are derived from known premises' and this standard includes an AI-specific note stating: '[i]n AI, a premise is either a fact, a rule, a model, a feature or raw data."
- Refers to the ability of the AI system, predominantly in the 'use phase', to generate outputs based on inputs. A 'capability of AI systems to derive models or algorithms, or both, from inputs or data' refers primarily, but is not limited to, the 'building phase' of the system and underlines the relevance of the techniques used for building a system.
- Machine learning approaches (Recital 12, AIA): supervised learning, unsupervised learning, self-supervised learning, reinforcement learning, deep learning
- Logic- and knowledge-based approaches: infer from encoded knowledge or symbolic representation of the task to be solved'. Instead of learning from data, these AI systems learn from knowledge including rules, facts and relationships encoded by human experts.

# AI definition in AI Act

## 6. Outputs that can influence physical or virtual environments:

- **Predictions** are one of the most common outputs that AI system produce and that require the least human involvement. A prediction is an estimate about an unknown value (the output) from known values supplied to the system (the input). Software systems have been used for decades to generate predictions. AI systems using machine learning are capable of generating predictions that uncover complex patterns in data;
- **Content** refers to the generation of new material by an AI system. This may include text, images, videos, music and other forms of output. Content can be understood from a technical perspective as a sequence of 'predictions' or 'decisions';
- **Recommendations** refer to suggestions for specific actions, products, or services to users based on their preferences, behaviours, or other data inputs.
- **Decisions** refer to conclusions or choices made by a system. An AI system that outputs a decision automates processes that are traditionally handled by human judgement

# AI definition in AI Act

## 7. Systems outside the scope of the AI system definition:

- **Systems for improving mathematical optimization**
  - systems used to improve mathematical optimisation or to accelerate and approximate traditional, well established optimisation methods. While those models have the capacity to infer, they do not transcend 'basic data processing'.
- **Basic data processing**
  - refers to a system that follows predefined, explicit instructions or operations. These systems are developed and deployed to execute tasks based on manual inputs or rules, without any 'learning, reasoning or modelling' at any stage of the system lifecycle. They operate based on fixed human-programmed rules, without using AI techniques.
- **Systems based on classical heuristics**
  - problem-solving techniques that rely on experience-based methods to find approximate solutions efficiently. Heuristics techniques are commonly used in programming situations where finding an exact solution is impractical due to time or resource constraints.
- **Simple prediction systems**
  - performance can be achieved via a basic statistical learning rule. financial forecasting (basic benchmarking) such machine-based systems may be used to predict future stock prices by using an estimator with the 'mean' strategy to establish a baseline prediction (e.g., always predicting the historical average price).

# AI definition in AI Act

JOURNAL ARTICLE

## In Search of the Lost Research Exemption: Reflections on the AI Act

Michèle Finck    Author Notes

*GRUR International*, Volume 74, Issue 10, October 2025, Pages 903–904, https://doi.org/10.1093/grurint/ikaf100

**Published:** 19 August 2025

- Art. 2(8) any research, testing or development activity prior to their being placed on the market or put into service
- Art. 2(6) specifically developed and put into service for the sole purpose of scientific research and development

# Questions?

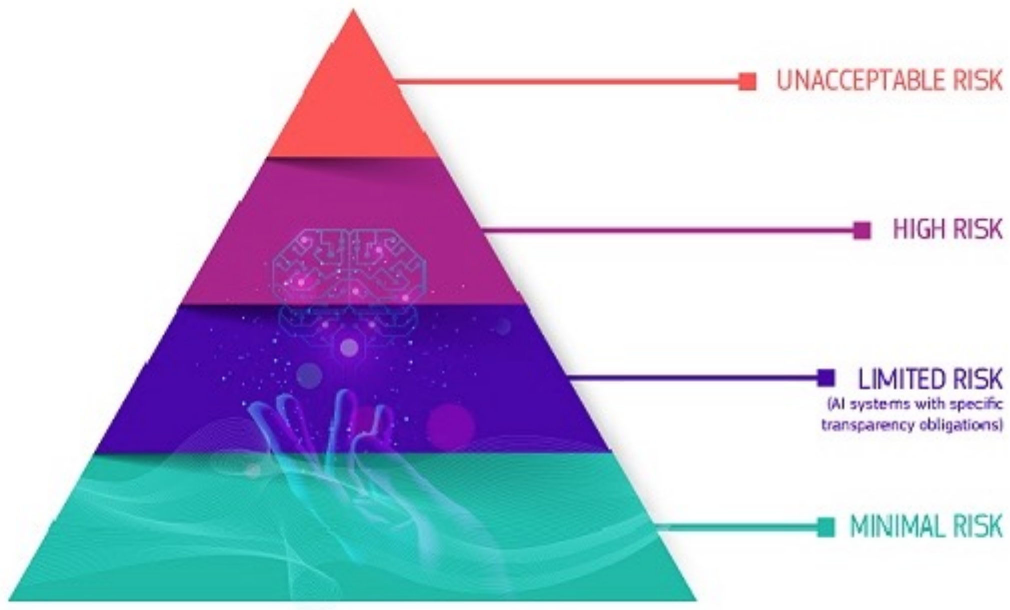
How does the definition of AI in the EU AI Act differ from your expectation?

# Part III – Overview of Risk-based Approach in EU AI Act

# Overview of Risk-based Approach in EU AI Act

## A risk-based approach

The AI Act defines 4 levels of risk for AI systems:




- Splits AI into four different bands of risk based on the intended use of a system;
- No sliding scale of risk, merely one category (‘high risk’)
- Some minor transparency provisions for a small number of systems characterised as ‘limited-risk’ AI;

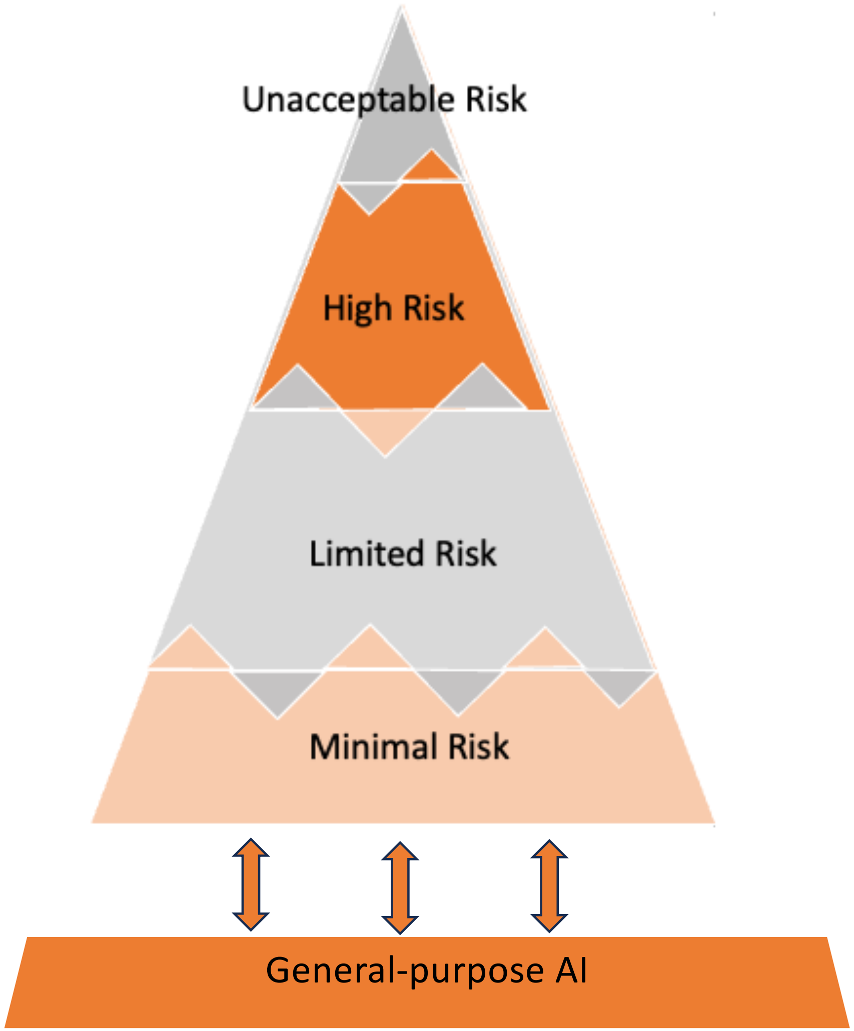
# Overview of Risk-based Approach in EU AI Act

- **Types of AI in AIA:**
  - Unacceptable Risk: e.g., Manipulative or Deceptive Techniques
  - High Risk: Biometric Identification, Education
  - Limited Risk: AI Systems Interacting Directly with Humans
  - Minimal Risk: AI-enabled video games

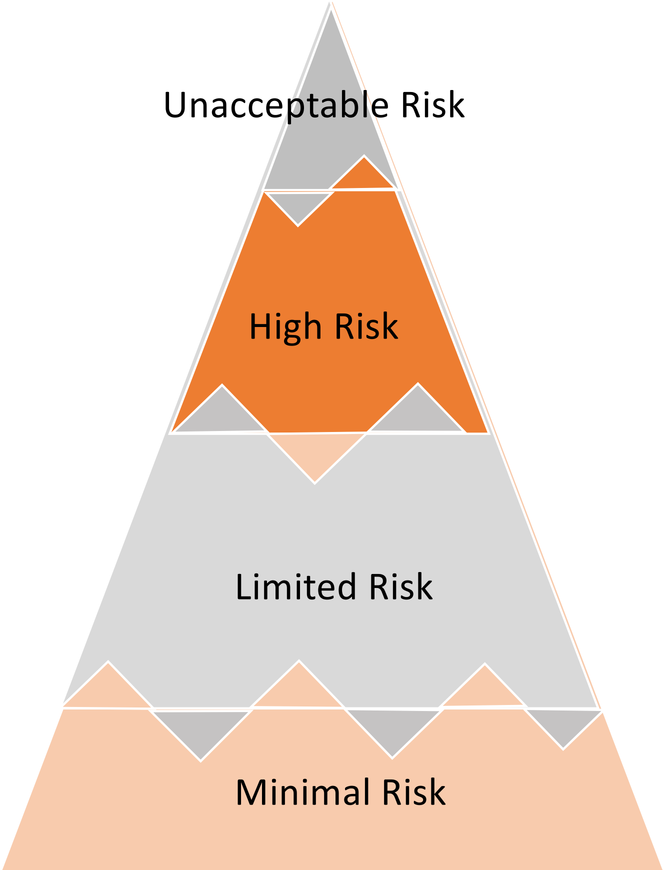
- AI types in AIA is not technological AI types
  - Overlap of AI Types: A single AI technology can fall into different risk categories based on its application.
  - For instance, facial recognition technology could be considered high-risk when used for law enforcement purposes but may be categorized differently in other contexts.

- Application-Based Classification: The AI Act's risk categories are determined by the specific use cases and contexts in which AI systems are deployed, not by the underlying technology itself.

# Overview of Risk-based Approach in EU AI Act




- Add general-purpose AI in the last minutes
- Each different types of AI could overlap




| AI Act (Articles) | Content | Example | Obligation |
|---|---|---|---|
| Art.5<br>Prohibited AI Practices | Vilolation of EU fundamental rights and values | Biometric categorisation systems, | Prohibition |
| Art.6<br>Classification Rules for High-risk<br>AI Systems & Annex III | Impact on health, safety or fundamental rights | Medical devices | Art.9-22<br>Conformity assessment, post-market monitoring |
| Art.50<br>Transparency Obiligations for Providers and Deployers of Certain AI System | Risks of impersonation, manipulation or deception | Chatbots, deep fakes | Information and transparency obiligation |
| No specific article | Common AI system | Spam filters, recommendersy stem | No specific regulation |

# Questions?

# Part IV – Type of AI in AI Act

# Type of AI in AI Act

- **Unacceptable Risk - Prohibited AI Practices:**
  - **Manipulative or Deceptive Techniques**: AI systems that employ subliminal, manipulative, or deceptive methods beyond an individual's conscious awareness, aiming to significantly distort behaviour and impair the ability to make informed decisions, leading to actions causing or likely to cause significant harm.
    - Example: AI shopping: AI-driven application that subtly influences users to make purchases they would not have otherwise considered, resulting in financial loss.
  - **Exploitation of Vulnerabilities:** AI systems that exploit vulnerabilities of individuals or specific groups due to age, disability, or socio-economic conditions, with the intent or effect of significantly distorting behaviour in a harmful manner.
    - Example: An AI-based game targeting children, designed to exploit their developmental vulnerabilities, encouraging excessive in-game purchases.
  - **Social Scoring**: AI systems used to evaluate or classify individuals or groups over time based on social behaviour or personal characteristics, leading to detrimental or unfavourable treatment unrelated to the original context of data collection, or treatment that is unjustified or disproportionate.
    - Aggregates data on individuals' behaviours to assign social scores, then used to unjustly deny services or opportunities.

# Type of AI in AI Act

- **Unacceptable Risk - Prohibited AI Practices:**
  - **Predictive Policing Based on Profiling**: AI systems used to assess or predict the likelihood of individuals committing criminal offenses, based solely on profiling or assessing personality traits and characteristics, without objective and verifiable facts directly linked to criminal activity.
  - **Unauthorized Facial Recognition Database Creation:** AI systems that create or expand facial recognition databases through untargeted scraping of facial images from the internet or CCTV footage without consent.
  - **Emotion Recognition in Workplace and Education**: AI systems used to infer emotions of individuals in workplaces and educational institutions, except when used for medical or safety reasons.
  - **Biometric Categorization for Sensitive Attributes**: AI systems that categorize individuals based on biometric data to deduce or infer sensitive attributes such as race, political opinions, religious beliefs, or sexual orientation.
  - **Real-Time Remote Biometric Identification in Public Spaces for Law Enforcement**: The use of real-time remote biometric identification systems in publicly accessible spaces for law enforcement purposes, with specific exceptions where such use is strictly necessary for objectives like searching for specific victims or preventing imminent threats.

# Type of AI in AI Act

- **High-Risk AI Systems:**
  - **Biometric Identification and Categorization**: AI systems used for real-time and post-event biometric identification of individuals, including facial recognition technologies.
  - **Critical Infrastructure Management:** AI applications in managing critical infrastructure, such as energy, transportation, and water supply, where failures could endanger lives.
  - **Education and Vocational Training**: AI systems determining access to education or evaluating students, like those used in admission processes or grading.
  - **Employment and Worker Management:** AI tools used in hiring processes, promotions, task assignments, or monitoring employee performance.
  - **Access to Essential Services**: AI systems assessing creditworthiness or determining access to essential public services, such as social benefits.
  - **Law Enforcement**: AI applications used by law enforcement for risk assessments, profiling, or predicting criminal activities.
  - **Migration and Border Control**: AI systems employed in managing migration, asylum applications, or border control, including lie detection tools.
  - **Administration of Justice**: AI tools assisting judicial authorities in interpreting laws and making decisions.

# Type of AI in AI Act

- **Limited-risk AI systems:**
  - **AI Systems Interacting Directly with Humans**:
    - Chatbots, virtual assistants, and customer service AI.
    - Obligations: Providers must ensure users are aware they are interacting with AI, unless it is obvious to a "reasonably well-informed" person. Exceptions apply to AI used for law enforcement (e.g., crime detection) with safeguards for rights
  - **AI Generating Synthetic Content**
    - Systems producing synthetic audio, images, videos (e.g., deepfakes), or text (e.g., ChatGPT).
    - Provider Obligations: Outputs must be machine-readable and detectable as AI-generated. Technical solutions must be robust, interoperable, and state-of-the-art, balancing feasibility and cost 1515.
    - Exceptions: Standard editing tools or systems that do not substantially alter input data
  - **AI-Generated Text for Public Interest**
    - Examples: News articles, public announcements.
    - Deployer Obligations: Disclosure is required unless the content undergoes human editorial review or is used for law enforcement

# Type of AI in AI Act

- **Limited-risk AI systems:**
  - **Exemptions and Key Exceptions**
    - **Law Enforcement**: Transparency obligations do not apply to AI authorized for crime detection, prevention, or prosecution.
    - **Creative Works**: Deepfakes in artistic/satirical contexts require minimal disclosure (e.g., a general notice).
    - **Human Oversight**: AI-generated text in public interest is exempt if humans hold editorial responsibility

# Type of AI in AI Act

- **Minimal risk AI:**
  - Minimal risk AI systems are those that do not fall into the unacceptable, high-risk, or limited-risk categories. They are considered safe and non-manipulative, with no significant potential to harm individuals or violate fundamental right
  - Examples include:
    - Spam filters
    - AI-enabled video games (e.g., NPC behaviour algorithms)
    - Inventory management tools
    - Ad-blocking software
    - Basic recommendation systems (e.g., non-personalized product suggestions)
  - No Mandatory Obligations

  Unlike higher-risk categories, minimal risk AI systems:
    - Do not require conformity assessments or registration in EU databases.
    - Need not comply with transparency or data governance rules (e.g., no need to inform users about AI interactions)

Codes of Conduct for Voluntary Application

# Questions?

# Part V – General-purpose definition in EU AI Act

# General-purpose AI definition in AI Act

## The scope of the EU AI Act

- Art. 3(63) ‘**general-purpose AI model**’ means an AI model, including where such an AI model is trained with **a large amount of data** using self-supervision at scale, that displays significant **generality** and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications, **except AI models that are used for research, development or prototyping activities before they are placed on the market**;
- Art. 3(66) ‘**general-purpose AI system**’ means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems。
- Art. 51 General-purpose AI model with **systemic risks**:
  - it has high impact capabilities evaluated on the basis of appropriate technical tools and methodologies, including indicators and benchmarks;
  - cumulative amount of computation used for its training measured in floating point operations is greater than 10(^25).

# General-purpose AI definition in AI Act

## The scope of the EU AI Act

➢ Annex XIII: the model has capabilities or high impact:

- the number of **parameters** of the model;
- the quality or size of the **data set**, for example measured through tokens;
- the amount of computation used for training the model, measured in **floating point operations** or indicated by a combination of other variables such as estimated cost of training, estimated time required for the training, or estimated energy consumption for the training;
- the benchmarks and evaluations of **capabilities** of the model, including considering the number of tasks without additional training, adaptability to learn new, distinct tasks, its level of **autonomy** and **scalability**, the **tools** it has access to;
- whether it has a high impact on the internal market due to its reach, which shall be presumed when it has been made available to **at least 10,000 registered business users** established in the Union;
- the number of registered end-users.

# Part V – Group Activity

# Group Discussion

Please analyse which risk category under the AI Act the following four AI companies should fall into?

**1. "EmoScan"**

Analyses facial expressions and voice tones to assess customer satisfaction during video calls.

**2. "EduBot"**

An AI tutor that adapts educational content while tracking student behaviour (e.g., eye movements, keystrokes).

**3. "DeepArt"**

Generative AI that creates hyper-realistic images, including replicas of living people.

**4. "CustomerHelper"**

Chatbot pretends to be human to provide customer service

*Should the risk classification focus on the tool's design or its potential misuse?*

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# THANK YOU!

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